Hongcheng Jiang

Applied AI Engineer | LLM Agents & Evaluation

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Kansas City, MO | Open to relocation / remote | U.S. Permanent Resident (no sponsorship required)

Summary

Applied AI engineer with 8 peer-reviewed publications, two tool-using agent projects, and a Ph.D. in Electrical and Computer Engineering. Builds and evaluates agents for SQL analysis and medical measurement, with bounded execution, failure recovery, and human review.

Technical Skills

Agents: Tool Calling, MCP, Structured Outputs, Bounded Repair, Evaluation, Execution Tracing
Engineering: Python, SQL, FastAPI, SQLite, Async Workers, Docker, Linux, Git
ML: PyTorch, Hugging Face, vLLM, Ollama, Computer Vision, Diffusion Models, Vision Transformers

Professional Experience

University of Missouri-Kansas City — Jun 2026 - Present

Research Associate (Postdoctoral Researcher) — *Kansas City, MO*

- RadMeasure — Medical Measurement Agent: Constrained LLM-proposed actions to registered protocols and tools; implemented geometry verification and a **KEEP/REPAIR/STOP controller** with repair budgets, independent-proposal checks, human review, and execution traces.
- RadMeasure repair evaluation: Across 528 runs on 176 archived cases, intervened in 106 runs (20.1%) with **0.69° mean error reduction among intervened runs**; overall MAE fell from 2.68° to 2.54°.
- GeoVIRA (submitted to WACV 2027): Trained geometry-conditioned models over **frozen medical image features** for angle estimation; compared cross-attention and measurement-aligned sampling using five-fold out-of-fold landmark predictions, three seeds, and matched-versus-shuffled controls.

NextTier – IT Solutions Consultancy — Jun 2025 - May 2026

Machine Learning Engineer — *Sacramento, CA (Remote)*

- Built SQL-Agent, a **Qwen3-14B natural-language SQL agent** for registered SQLite data, with browser and FastAPI/MCP interfaces for revenue totals, order lists, grouped summaries, and result review.
- Engineered schema-aware **generate–execute–repair** loops with read-only authorization, query/output limits, and up to three attempts on a consistent data snapshot; routed general answers to human review.
- Built durable asynchronous execution with idempotent jobs, renewable worker leases, retries, and **stale-worker write protection**; validated recovery using worker-kill, restart, duplicate-request, and timeout tests.
- Achieved **91.0% correct runs (142/156) versus 65.4% (102/156)** for the single-pass Coder baseline in a configuration comparison on 26 development questions × 2 databases × 3 trials; used asynchronous delivery for 74.8 s p95 latency.

University of Missouri-Kansas City — Jan 2020 - May 2025

Graduate Research Assistant — *Kansas City, MO*

- Developed and trained transformer- and diffusion-based models for **thermal super-resolution, hyperspectral pansharpening, and NIR-to-RGB translation**; evaluated image quality and spectral fidelity.
- Co-authored a fully connected LSTM-CRF for clinical named-entity recognition, achieving **84.15% F1 on i2b2/VA**.

Education

University of Missouri-Kansas City

Ph.D., Electrical & Computer Engineering, GPA: 4.0/4.0 — May 2025

M.S., Computer Science, GPA: 3.8/4.0 — Dec 2019

Publications

8 peer-reviewed publications; 6 first-author imaging papers. Full list: Google Scholar.